# CHAPTER SEVEN

## Memory organization

# MEMORY

- **Memory** is unit used for storage, and retrieval of data and instructions.
- A typical computer system is equipped with a hierarchy of memory subsystems, some internal to the system and some external.
- Internal memory systems are accessible by the CPU directly and
- external memory systems are accessible by the CPU through an I/O module.

# CONT..

- Memory systems are classified according to their the following characteristics

1. **Location:** The classification of memory is done according to the location of the memory as:

   - **Registers**: The CPU requires its own local memory in the form of registers and also control unit requires local memories which are fast accessible.
   - **Internal (main):** is often associated with the main memory (RAM)
   - **External (secondary):** consists of peripheral storage devices like Hard disks, magnetic tapes, etc.

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**2. Capacity:** Storage capacity is one of the important aspects of the memory.

- It is measured in bytes. Since the capacity of memory in a typical memory is very large, the prefixes kilo (K), mega (M), and giga(G).
- A kilobyte is $2^{10} = 1024$ bytes, a megabyte is $2^{20}$ bytes, and a giga byte is $2^{30}$ bytes.

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**3. Unit of Transfer:** Unit of transfer for internal memory is equal to the number of data lines into and out of memory module.

- **Word:** For internal memory, unit of transfer is equal to the number of data lines into and out of the memory module.
- **Block:** For external memory, data are often transferred in much larger units than a word, and these are referred to as blocks.

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## 4. Access Method

- **Sequential:** Tape units have sequential access. Data are generally stored in units called **"records"**.
  - Data is accessed sequentially;
- **Random:** Each addressable location in memory has a unique addressing mechanism. The time to access a given location is independent of the sequence of prior accesses and constant.
  - Any location can be selected at random and directly addressed and accessed. Main memory and cache systems are random access.

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## 5. Performance

- **Access time**: For random-access memory, this is the time it takes to perform a read or write operation:
- **Transfer rate**: This is the rate at which data can be transferred into or out of a memory unit.

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6. **Physical Type**

- **Semiconductor:** Main memory, cache. RAM, ROM.
- **Magnetic:** Magnetic disks (hard disks), magnetic tape units.
- **Optical:** CD, DVD.

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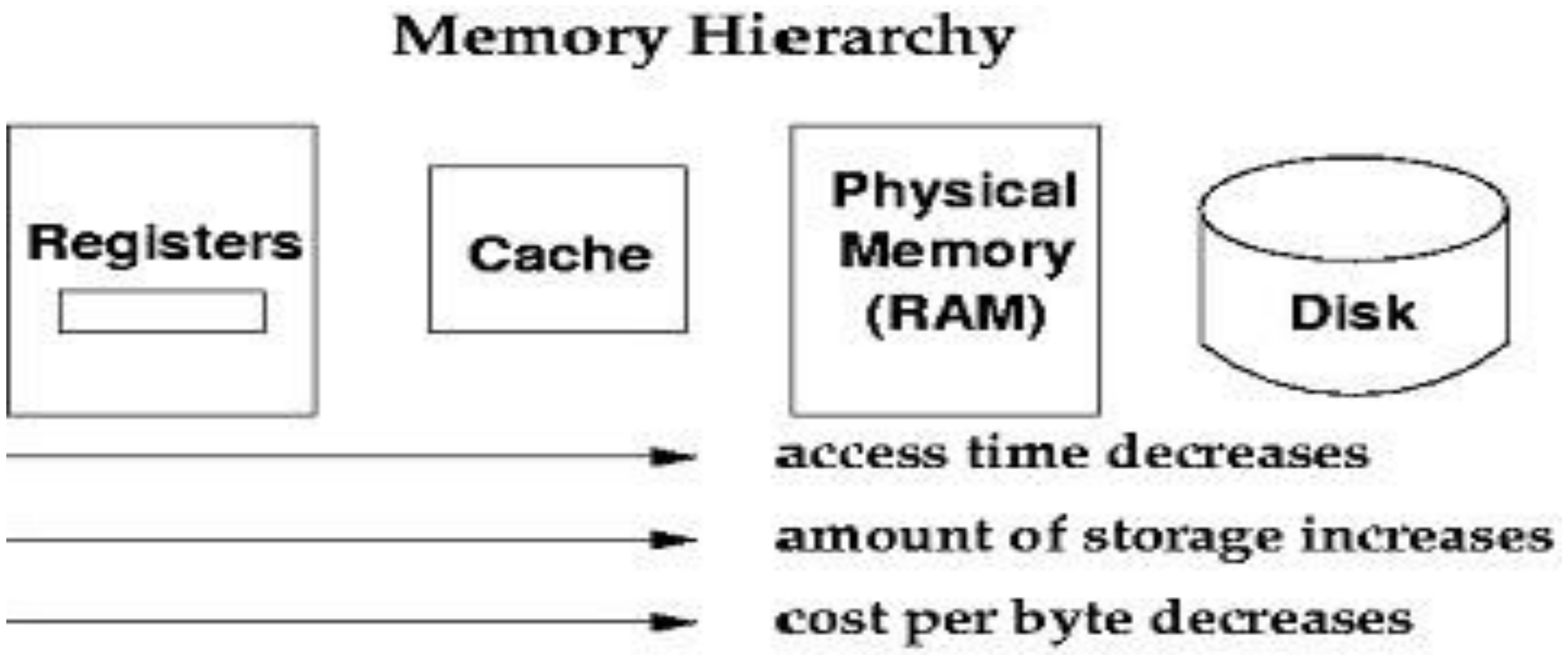
## 7. Physical Characteristics

- **Volatile/nonvolatile**: In a volatile memory, information decays naturally or is lost when electrical power is switched off.
- **Erasable/nonerasable:** Nonerasable memory cannot be altered (except by destroying the storage unit). ROM's are nonerasable.

# MEMORY HIERARCHY

- A computer system is equipped with a hierarchy of memory subsystems.
- There are several memory types with very different physical properties.
- The important characteristics of memory devices are cost per bit, access time, data transfer rate, alterability and compatibility with processor technologies.


Memory Hierarchy
Registers
Cache
Physical
Memory
(RAM)
Disk
access time decreases
amount of storage increases
cost per byte decreases

# TYPES OF MEMORIES(STORAGE DEVICES)

1. **Main Memory**

- The main memory (RAM) stores data and instructions In active use
- RAMs are built from semiconductor materials.
- Semiconductor memories fall into two categories,
  1. SRAMs (static RAMs)
  2. DRAMs (dynamic RAMs).

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I. **DYNAMIC RAM** (DRAM) is made with cells that store data as charge on capacitors.
   - The presence or absence of charge in a capacitor is interpreted as a binary 1 or 0.

II. **STATIC RAM** (SRAM)
   - binary values are stored using traditional flip-flop logic-gate.
   - A static RAM will hold its data as long as power is supplied to it.
   - Static RAM's are faster than dynamic RAM's.

# CONT..

- **ROM :** The data is actually wired in the factory.
  - The data can never be altered.
- **PROM:** Programmable ROM.
  - It can only be programmed once after its fabrication.
  - It requires special device to program.

# CONT..

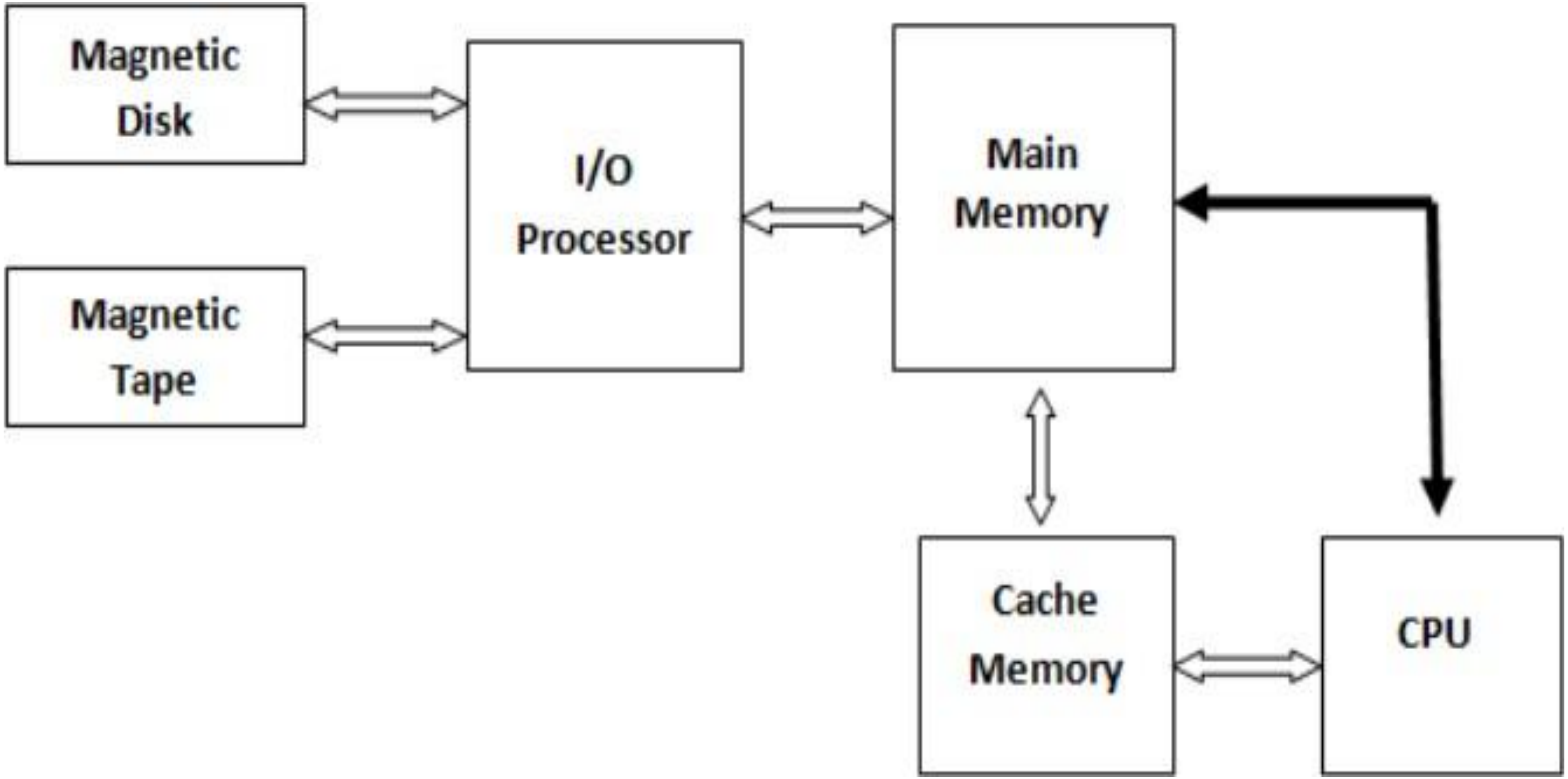
- **EPROM**: Erasable Programmable ROM.
  - It can be programmed multiple times.
  - Whole capacity need to be erased by ultraviolet radiation before a new programming activity.
  - It cannot be partially programmed.
- **EEPROM**: Electrically Erasable Programmable ROM.
  - Erased and programmed electrically.
  - It can be partially programmed.
  - Write operation takes considerably longer time compared to read operation.

# 2. CATCHE MEMORY

- **Cache memory** is a small, high-speed RAM buffer located between the CPU and main memory.
- **Cache memory** holds a copy of the instructions (instruction cache) or data (operand or data cache) currently being used by the CPU.
- The main purpose of a cache is to accelerate your computer while keeping the price of the computer low.

# CONT..

- High speed SRAM
- A memory that can be accessed more quicker than the regular main memory of Ram
- It is also called CPU Memory
- The performance of the cache memory is measured in terms of ***Hit Ratio***

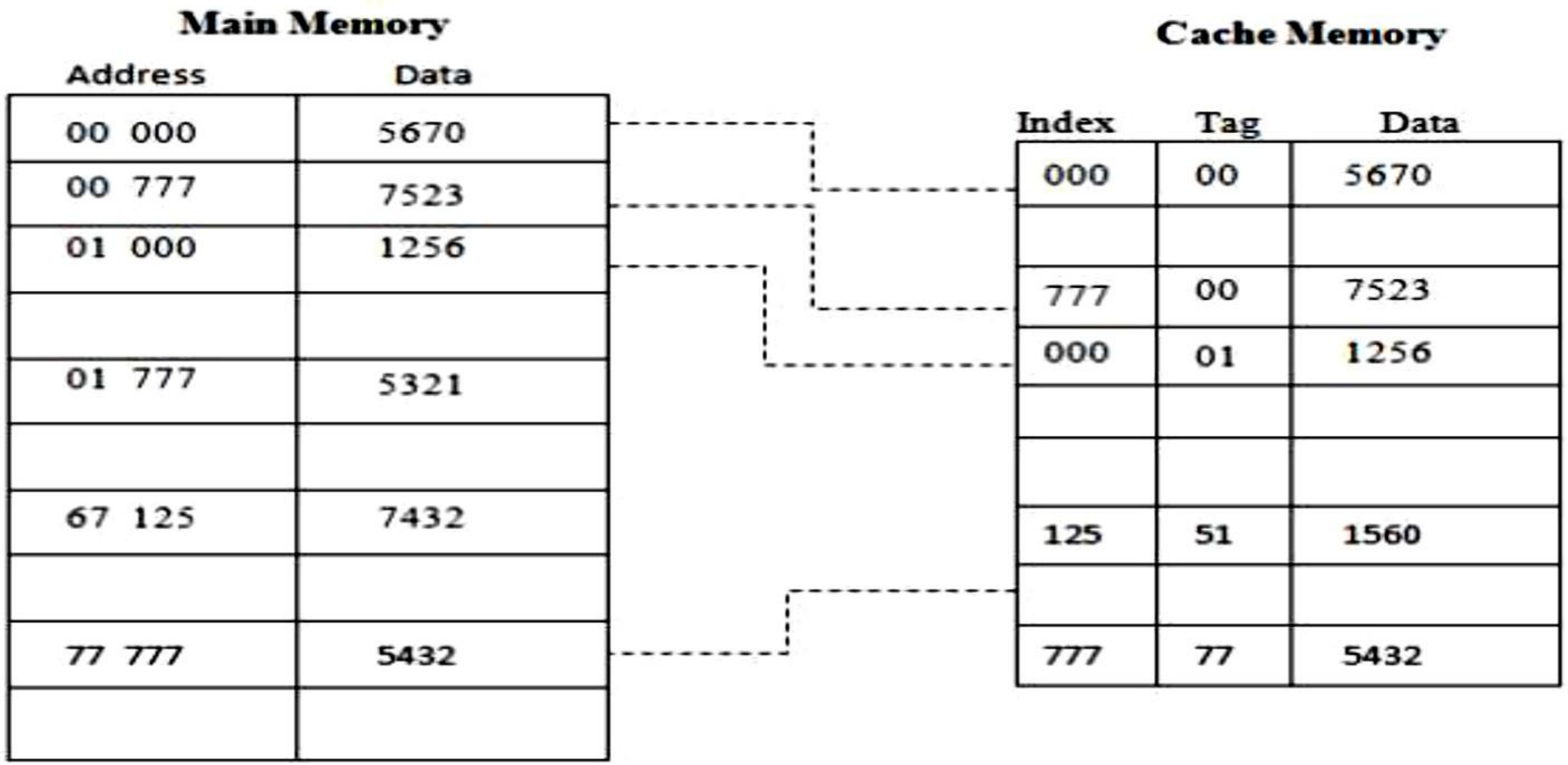

Magnetic Disk
Magnetic Tape
I/O Processor
Main Memory
Cache Memory
CPU

# CACHE MEMORY MAPPING

- Cache memory woks under three different configurations
    1. Direct mapped cache
    2. Fully associative cache
    3. Set associative cache mapping

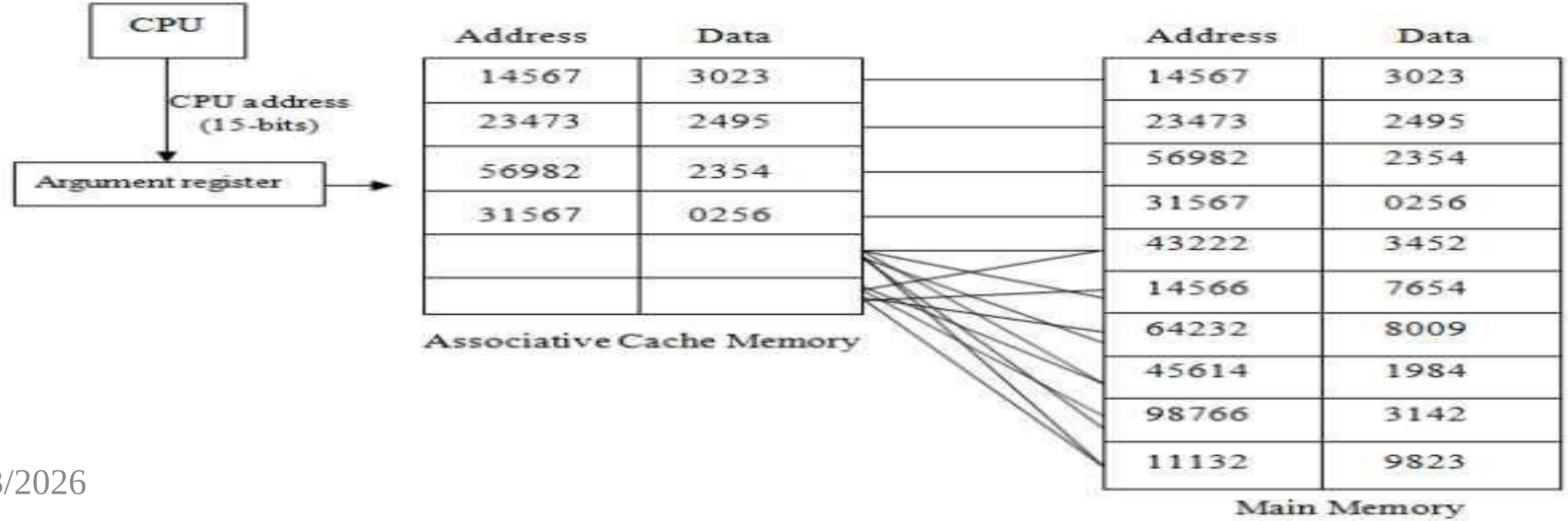
# 1. DIRECT MAPPED CACHE

- Has each block mapped to exactly one cache memory location.



# 2. ASSOCIATIVE MAPPING

- Similar to direct mapping in structure
- But allows block to be mapped to any Location rather than a predefined cache memory Location



# 3. SET ASSOCIATIVE MAPPING

- each cache location can have more than one pair of tag + data items.
- more flexible mapping of the fully associative cache.

# REPLACEMENT ALGORITHMS OF CACHE MEMORY

- Are used when there is no available space in a cache in which to place a data. Four of the most common algorithm.

1. **Least Recently Used (LRU):**
   - selects for replacement the item that has been least recently used by the CPU.
2. **First-In-First-Out (FIFO):**
   - selects for replacement the item that has been in the cache from the longest time.
3. **Least Frequently Used (LRU):**
   - The LRU algorithm selects for replacement the item that has been least frequently used by the CPU.
4. **Random:**
   - The random algorithm selects for replacement the item randomly.

# 3. VIRTUAL MEMORY

- The term **virtual memory** refers to something which appears to be present but actually it is not.
- The virtual memory technique allows users to use more memory for a program than the real memory of a computer.
- **virtual memory** is the concept that gives the illusion to the user that they will have main memory equal to the capacity of secondary storage media.

# CONCEPT OF VIRTUAL MEMORY

- A programmer can write a program which requires more memory space than the capacity of the main memory.
- Such a program is executed by virtual memory technique.
- The program is stored in the secondary memory.
- The memory management unit (MMU) transfers the currently needed part of the program from the secondary memory to the main memory for execution.
- The movement of instructions and data between the main memory and the secondary memory is called **Swapping**.

# 4. AUXILIARY MEMORY

- also referred to as *secondary storage*)
- *refer to as auxiliary storage, secondary storage, secondary memory, external storage or external memory*
- is the non-volatile memory lowest-cost, highest-capacity, and slowest-access storage in a computer system.
- It is where programs and data kept for long-term storage or when not in immediate use.

# CONT..

- **Auxiliary memory** is not directly accessible by the CPU;
  - large data files,
  - documents,
  - programs and
  - back up information that supplied to primary memory from auxiliary memory over a high-bandwidth channel, which will use whenever necessary.
- Auxiliary memory holds data for future use, and that retains information even the power fails.

# 5. Associative Memory

- A type of computer memory from which items may be retrieved by matching some part of their content, rather than by specifying their address.
- also called associative storage or *Content-addressable memory* (CAM).
- *Associative memory* is much slower than RAM, and is rarely encountered in mainstream computer designs.